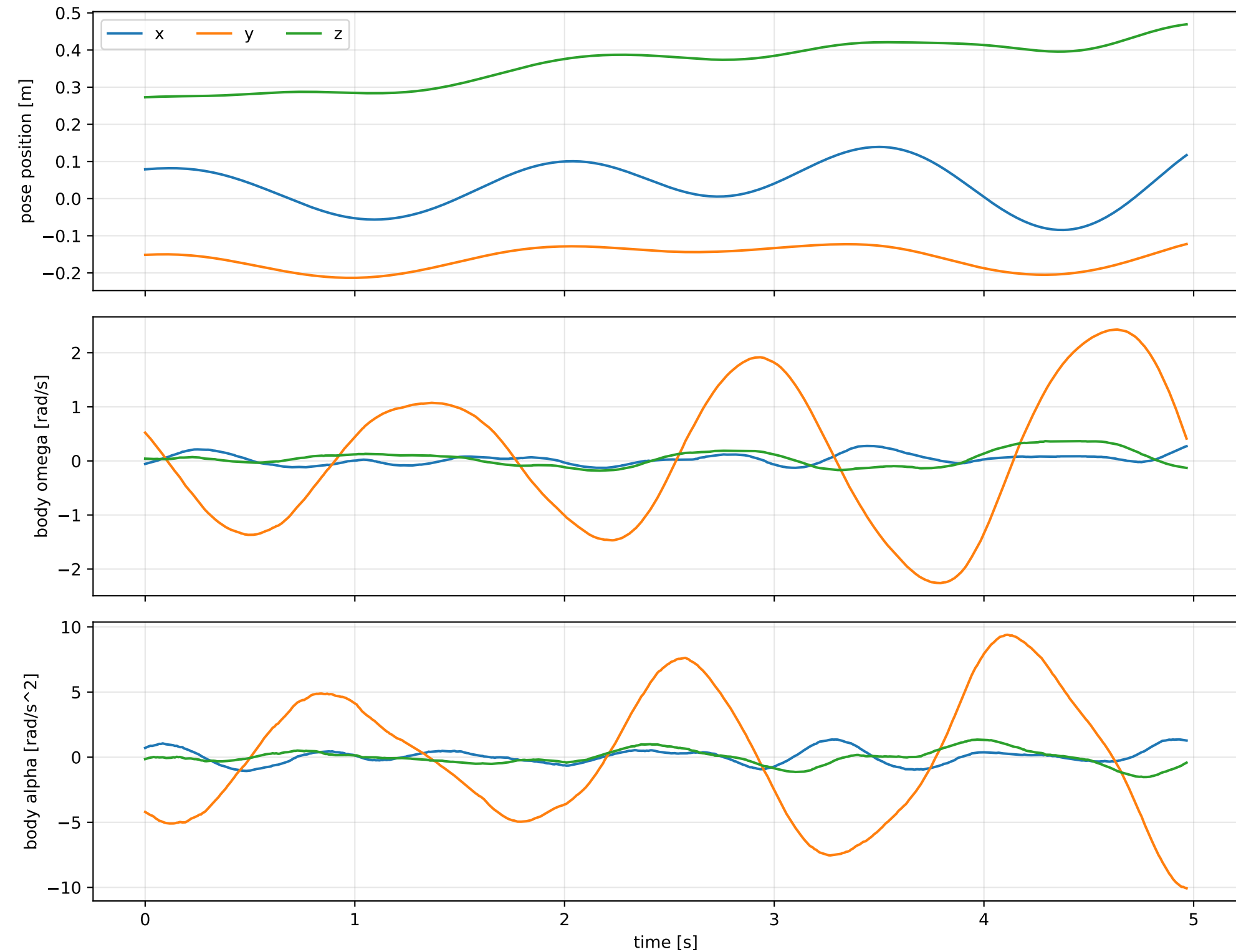
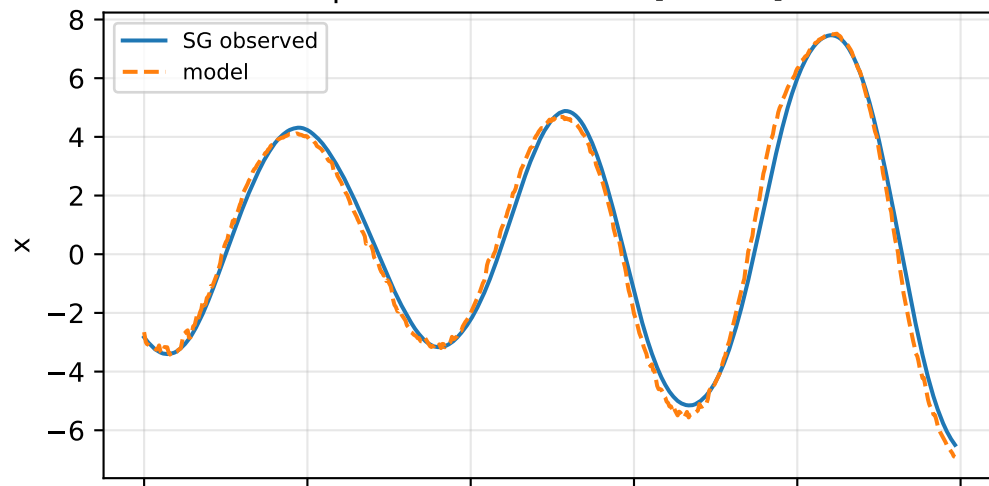


gimbal_command_replay: SG trajectory and derived motion

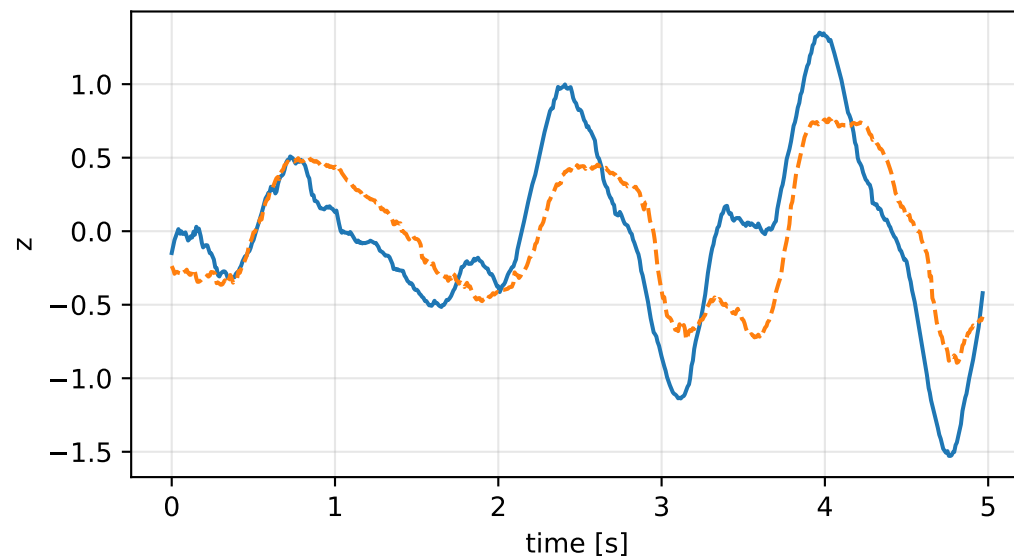
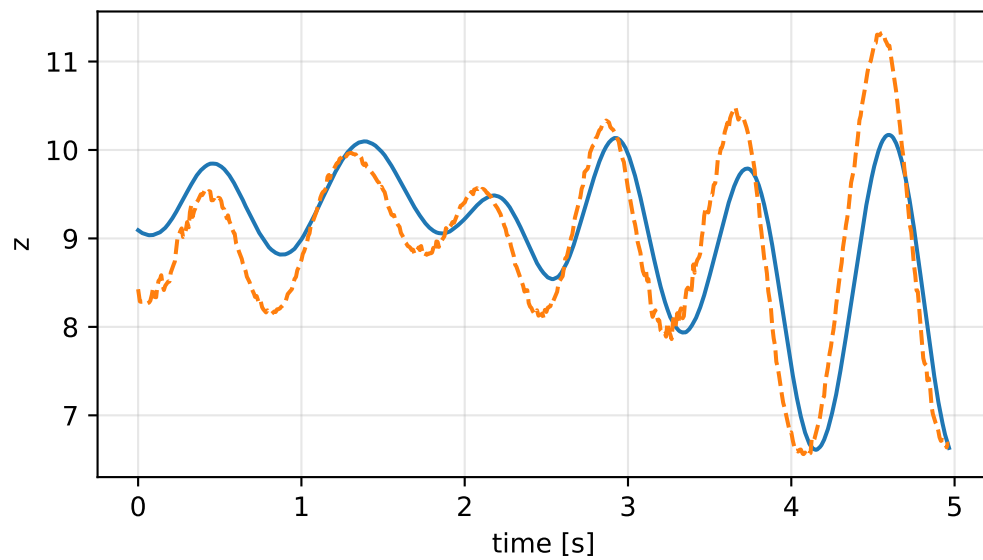
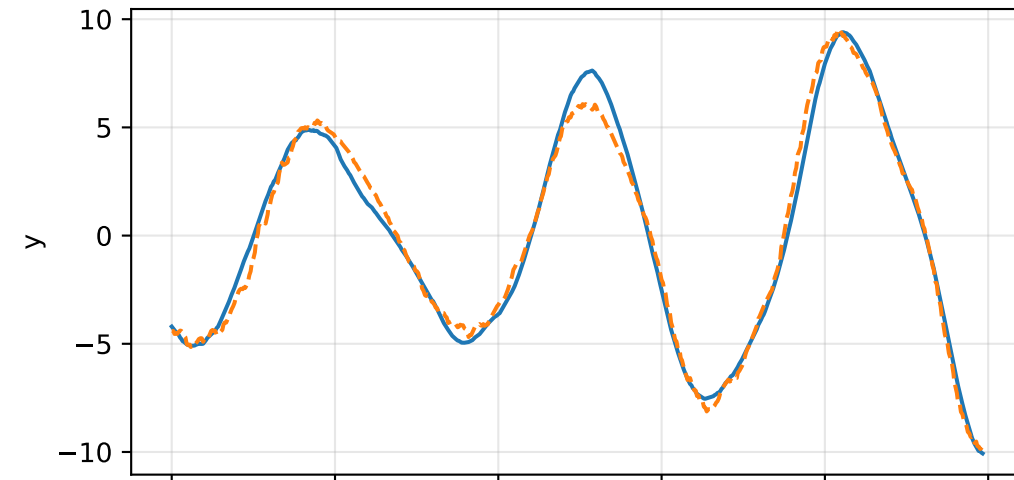
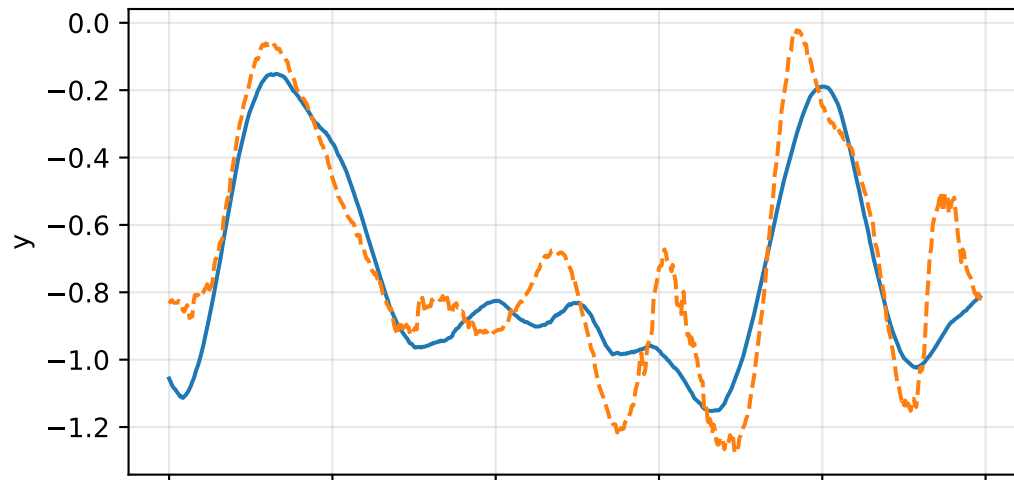
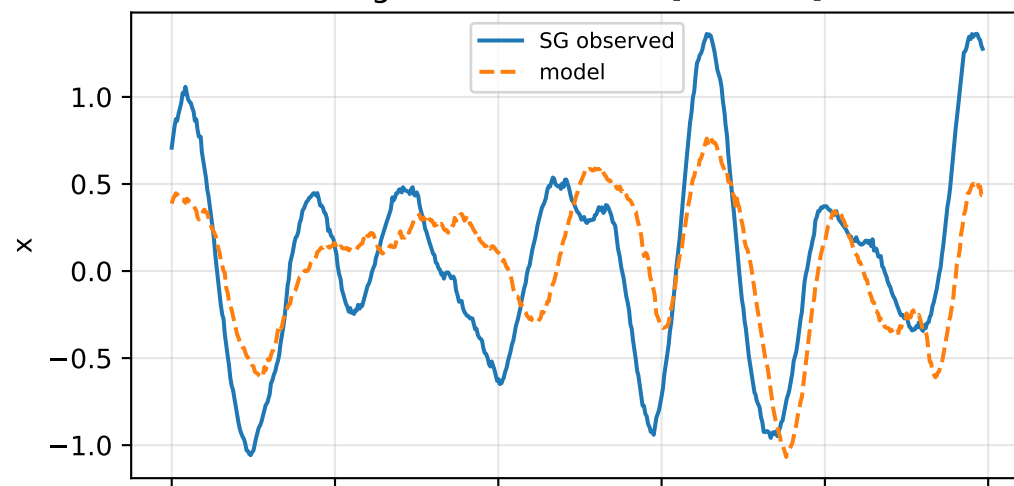


gimbal_command_replay: acceleration residual objective

specific acceleration [m/s^2]

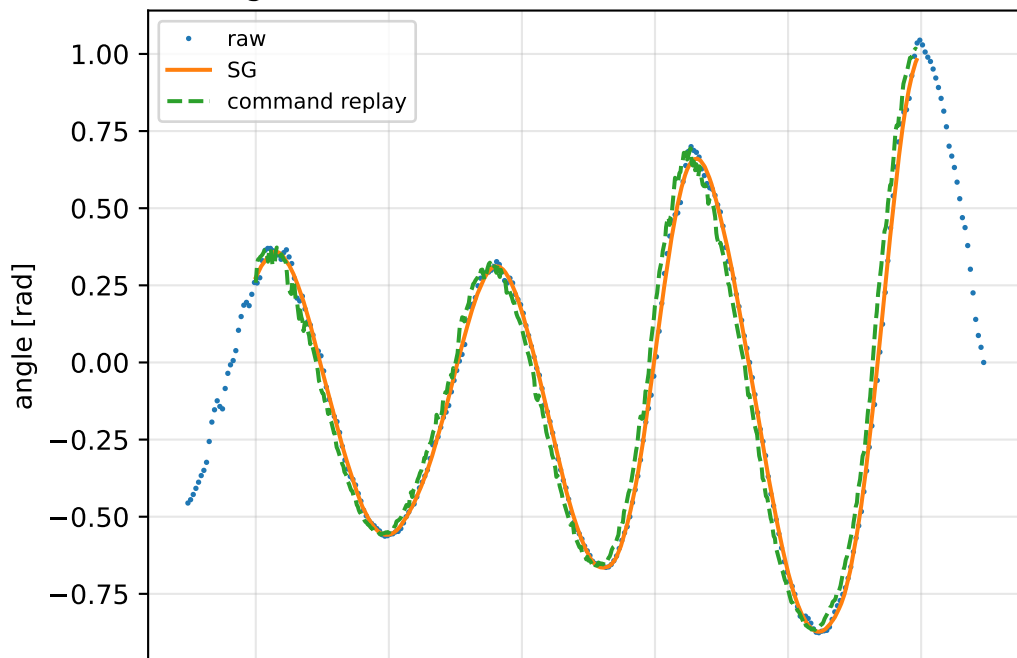


angular acceleration [rad/s^2]

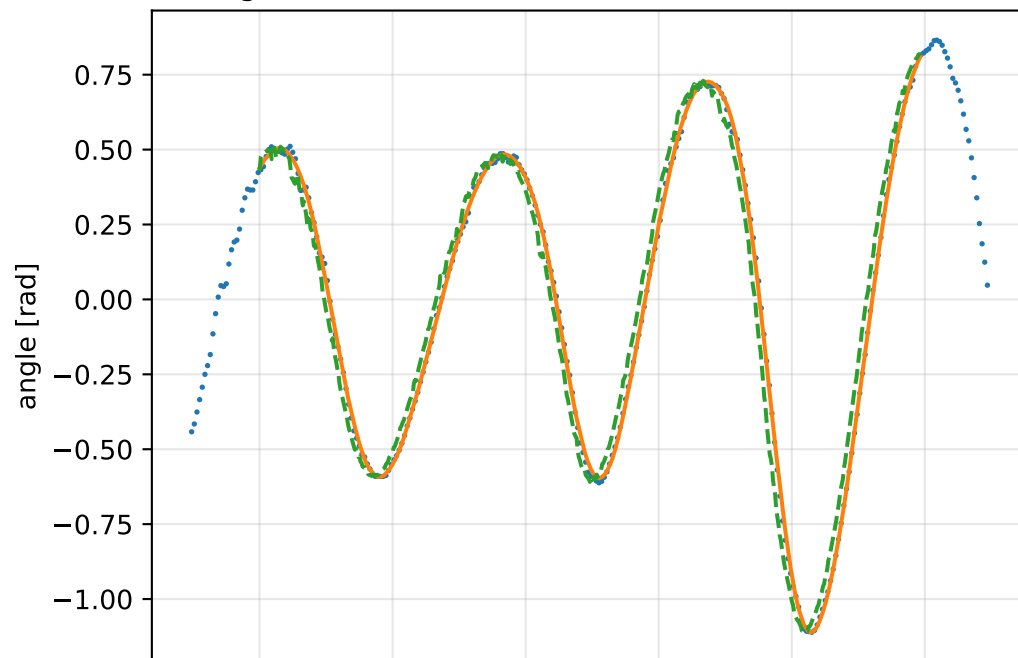


gimbal_command_replay: gimbal measurement audit (objective=command_replay)

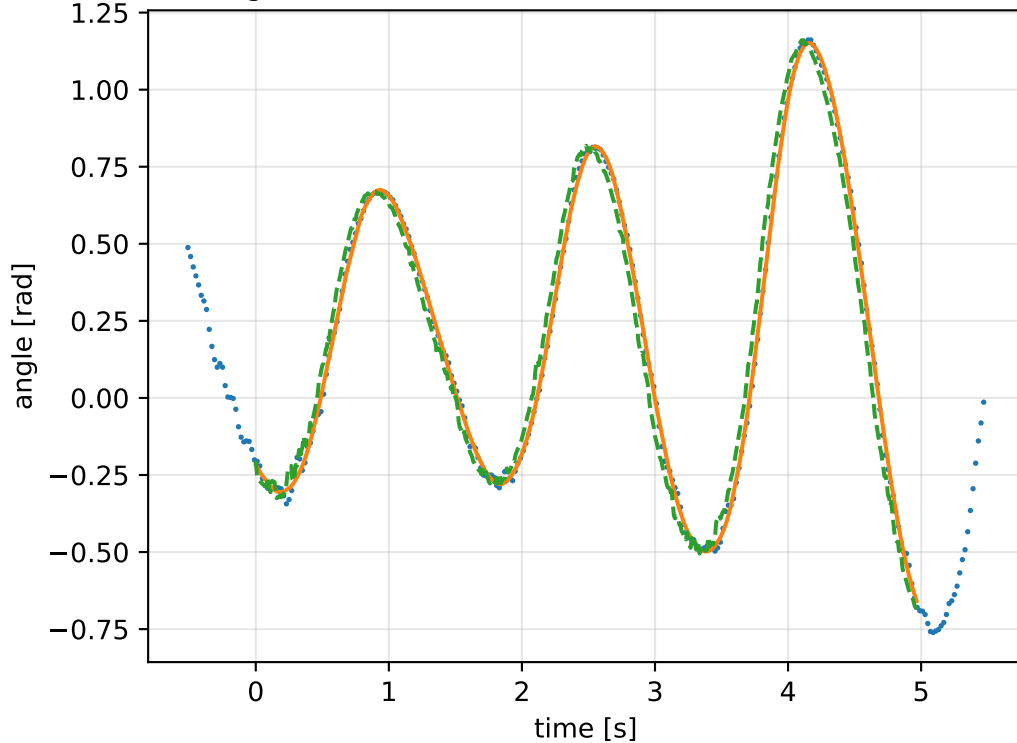
gimbal 1: RMSE=0.0759, max=0.2095 rad



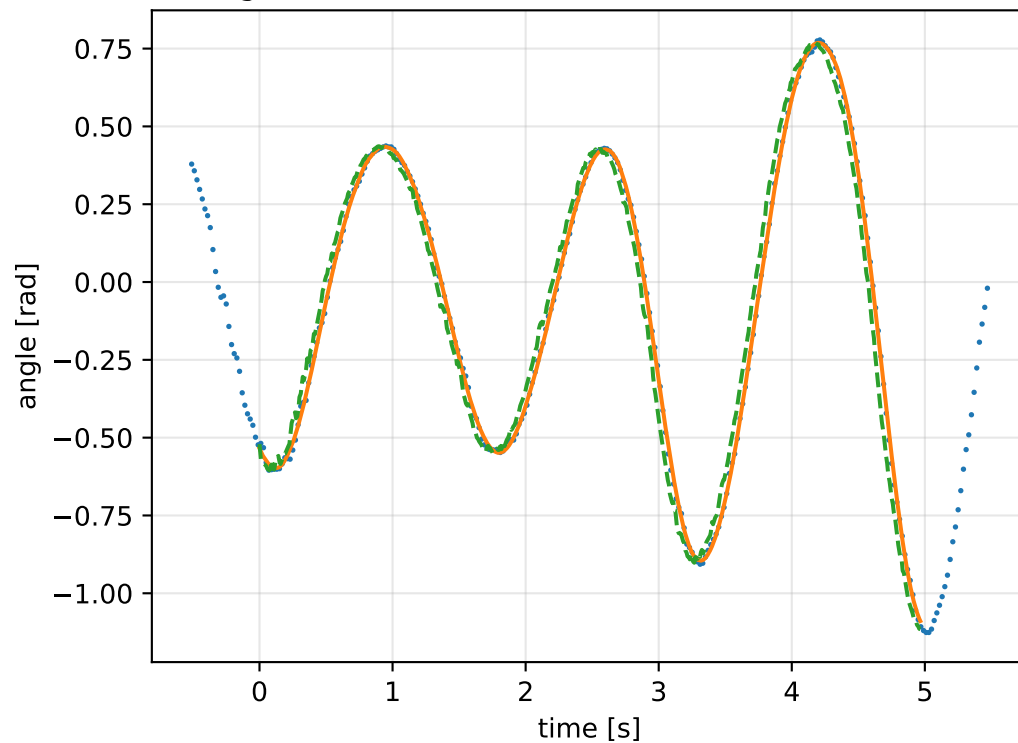
gimbal 2: RMSE=0.08099, max=0.2013 rad



gimbal 3: RMSE=0.07393, max=0.172 rad

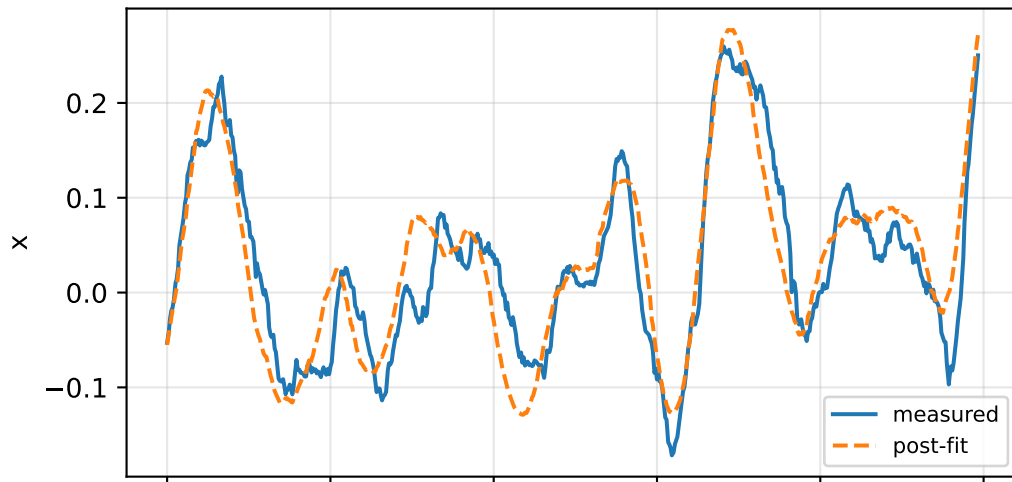


gimbal 4: RMSE=0.07131, max=0.1737 rad

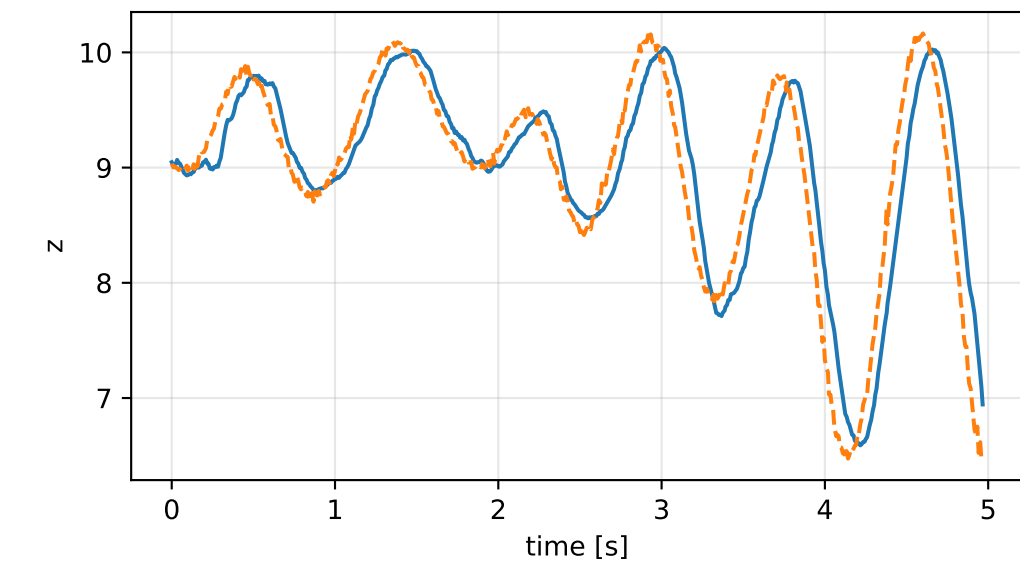
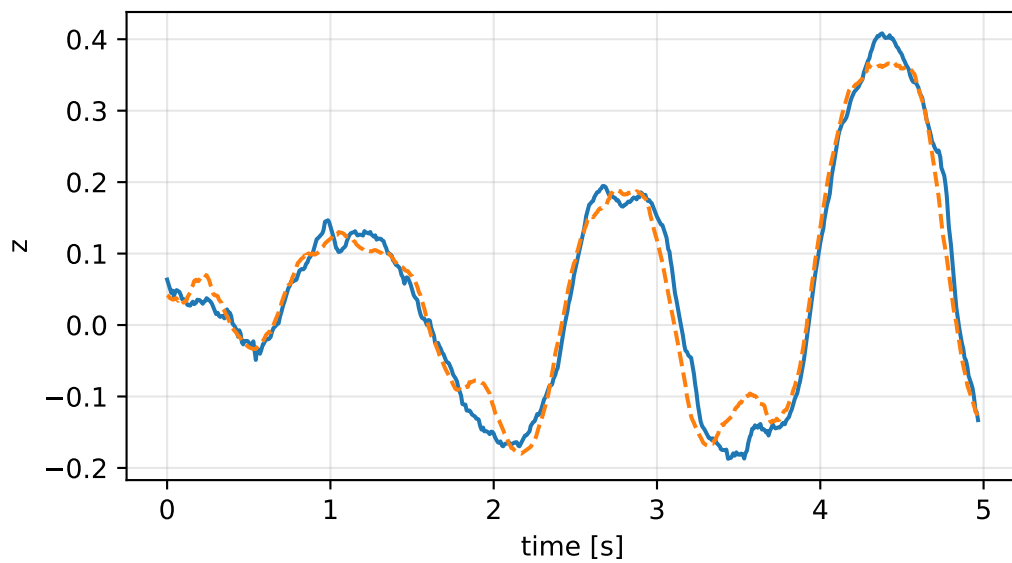
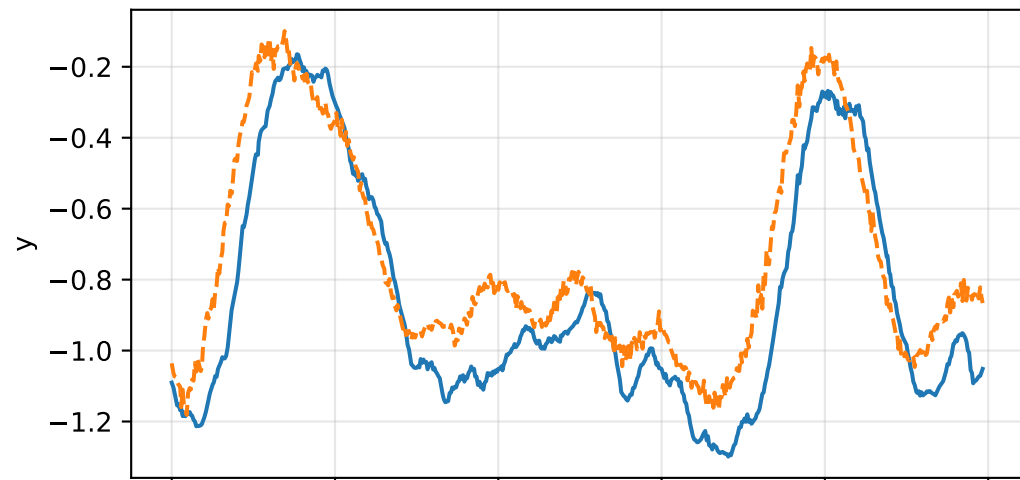
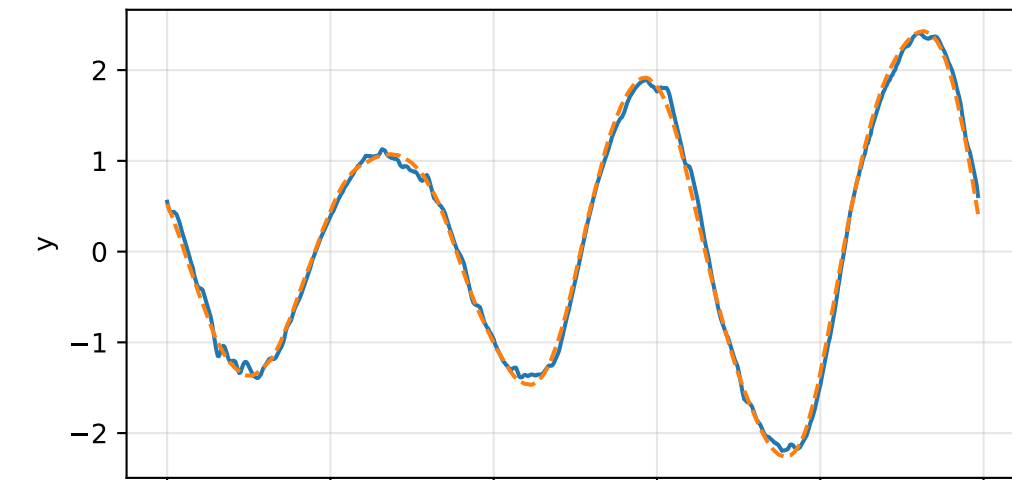
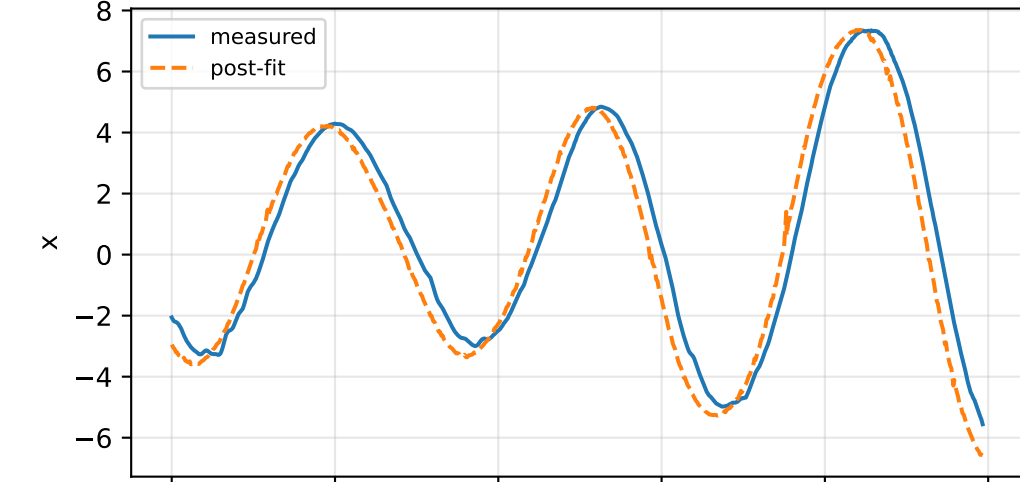


gimbal_command_replay: IMU comparison (diagnostic only)

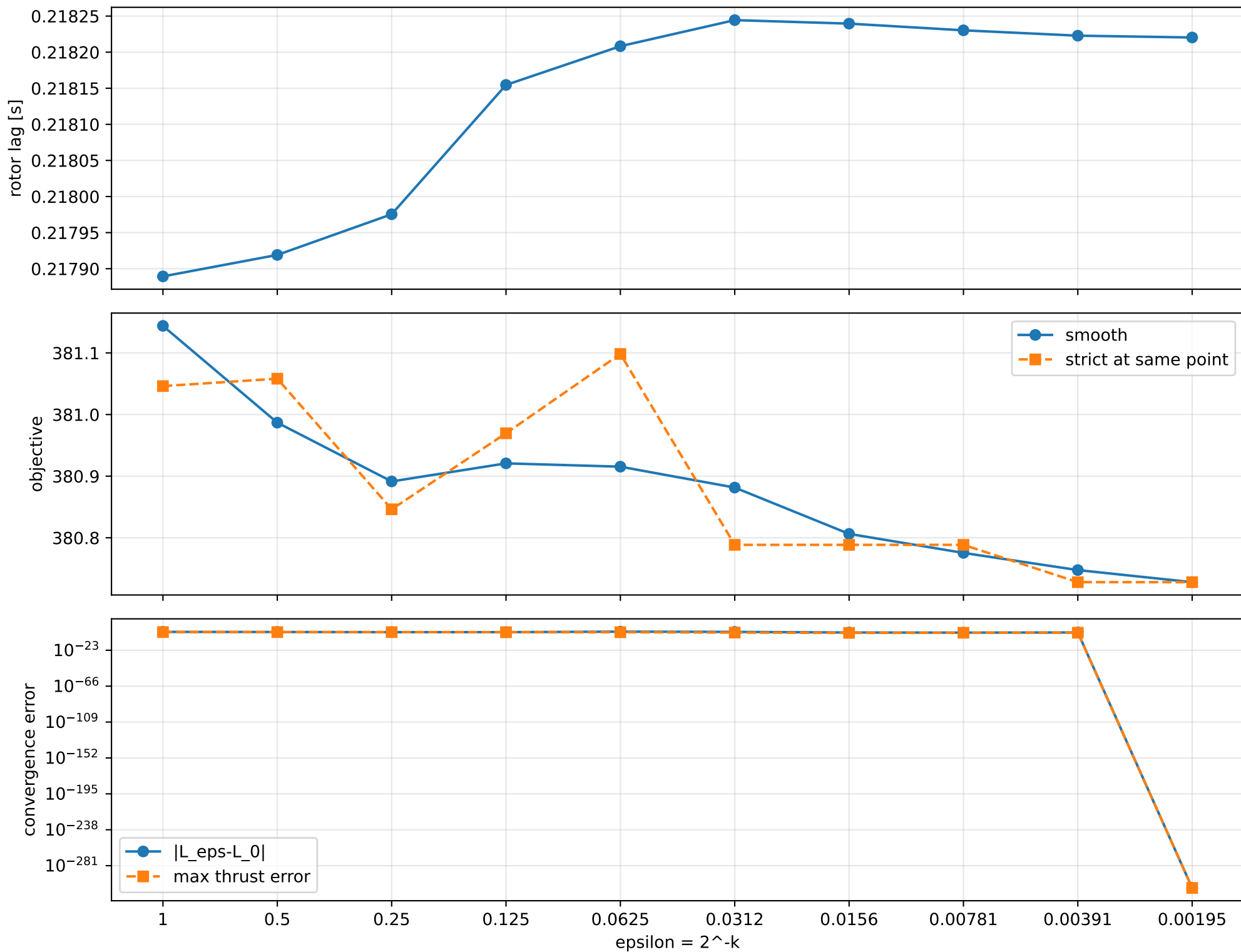
gyro [rad/s]



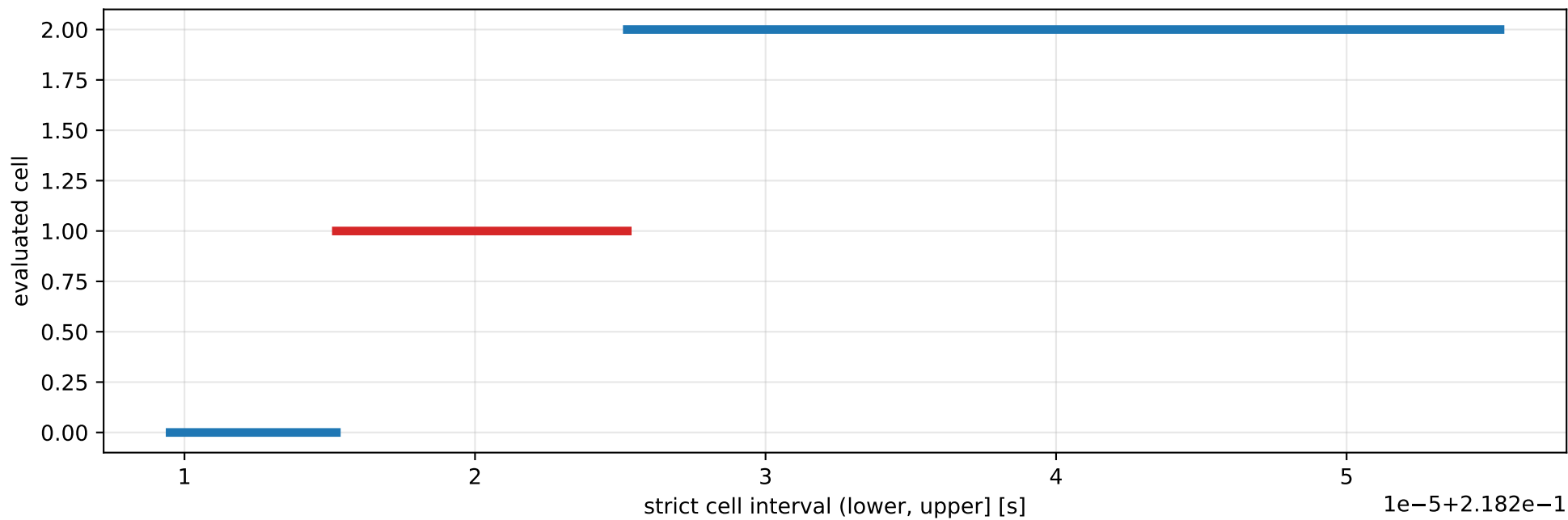
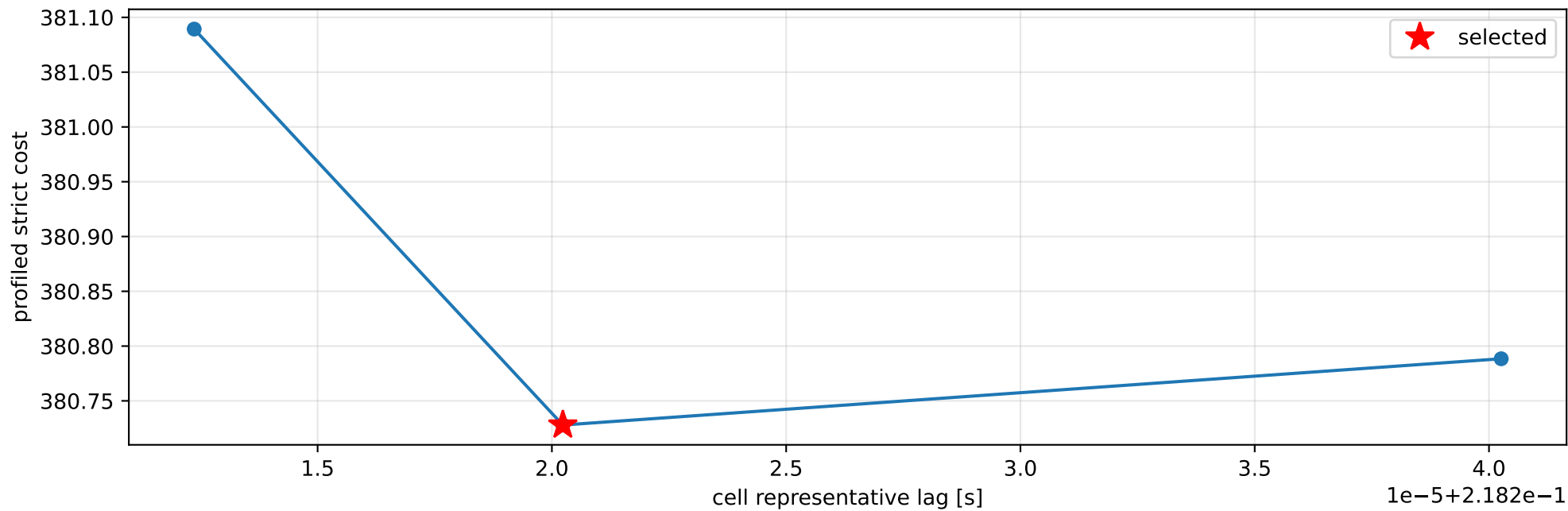
specific force [m/s^2]



gimbal_command_replay: smooth-to-strict continuation



gimbal_command_replay: exact strict-ZOH lag cells



selected (0.218215227, 0.218225241] s; final smooth 0.218220352 s; data boundary=False

gimbal_command_replay: parameter estimate

Case: gimbal_command_replay

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 4.15238489

inertia [kg m²]:

```
[[ 5.12078797e-01  5.34794957e-04 -1.02794807e-01]
 [ 5.34794957e-04  3.20318797e-01  3.49281649e-02]
 [-1.02794807e-01  3.49281649e-02  7.51810067e-01]]
```

CoG body [m]: [0.02433595 -0.02296906 -0.04706597]

force effectiveness: [1.45753571 1.95473746 1.4864644 1.14147787]

scale-free J/m [m²]:

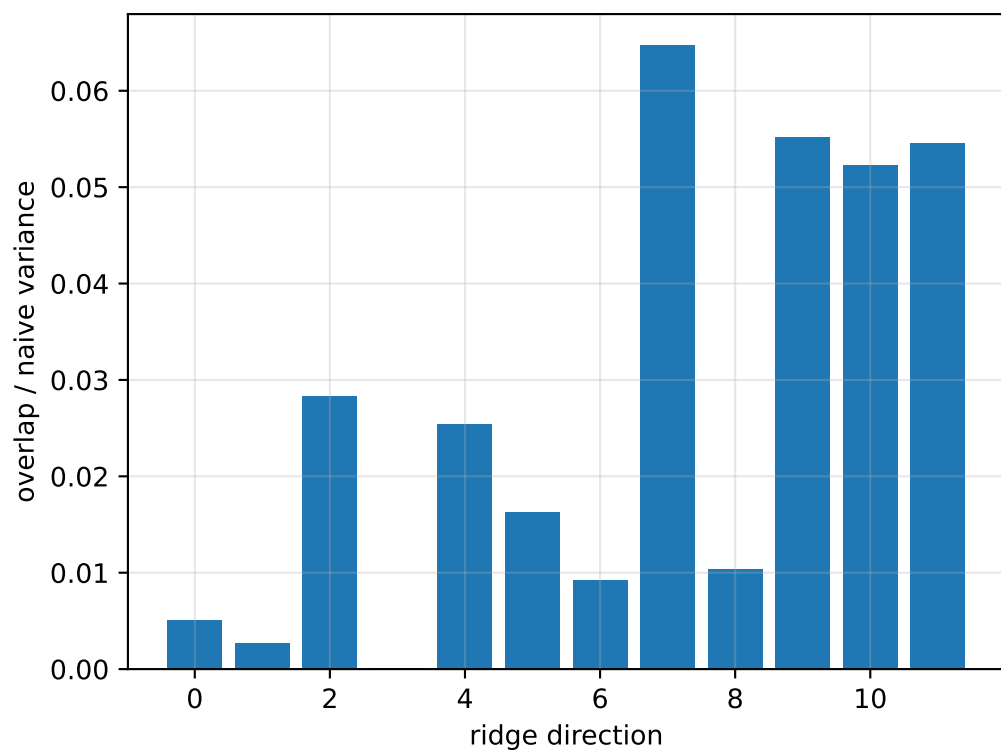
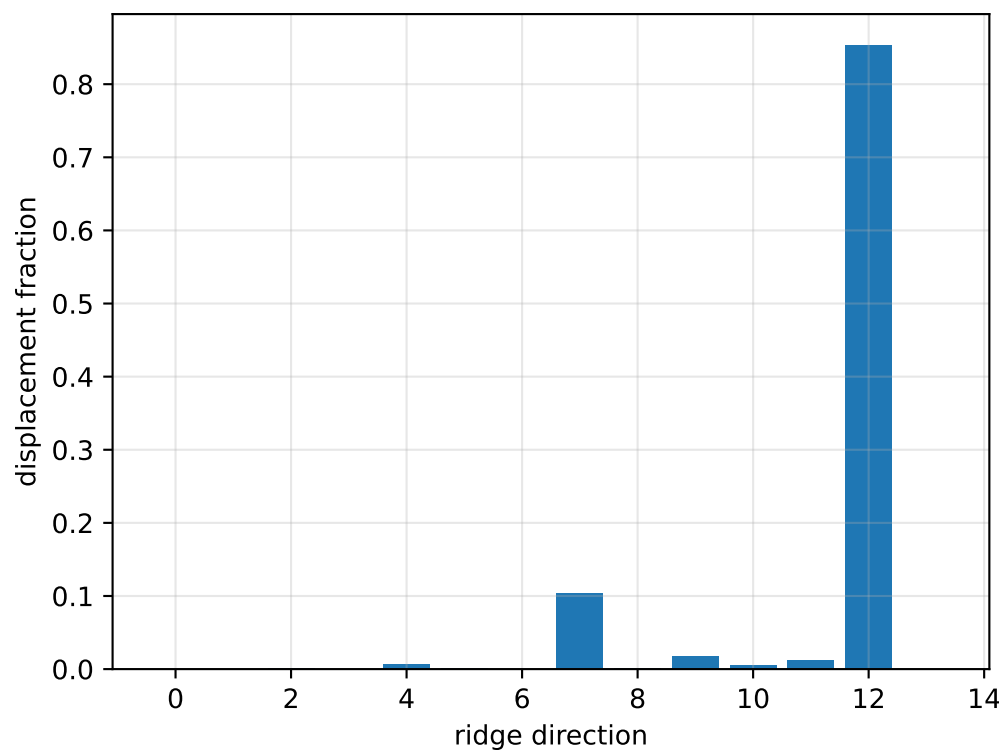
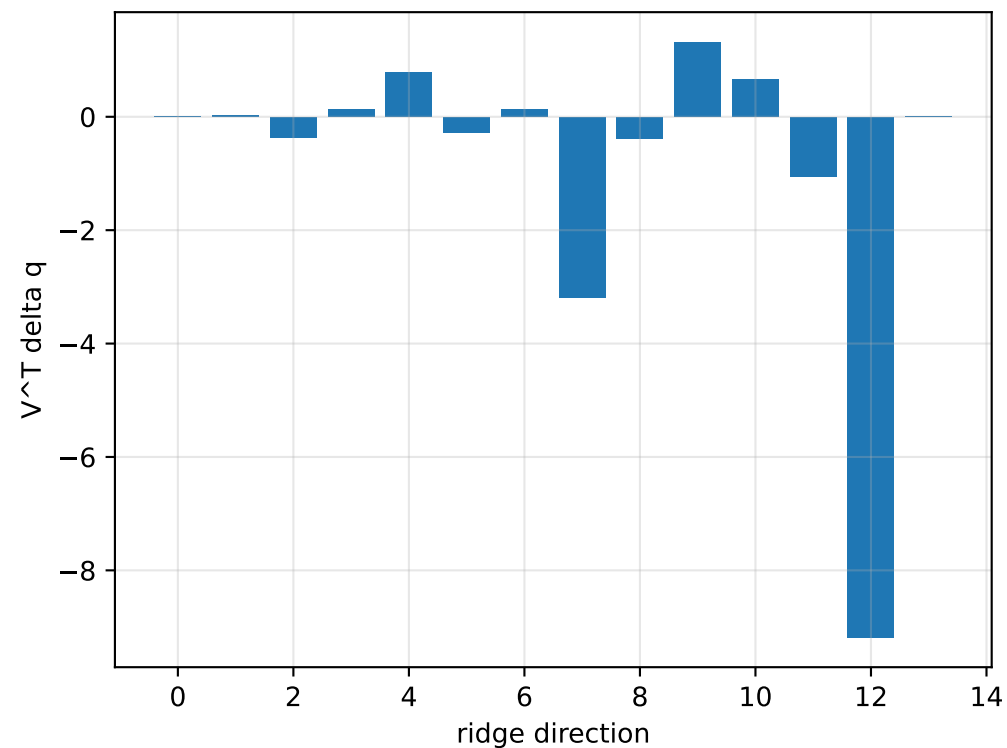
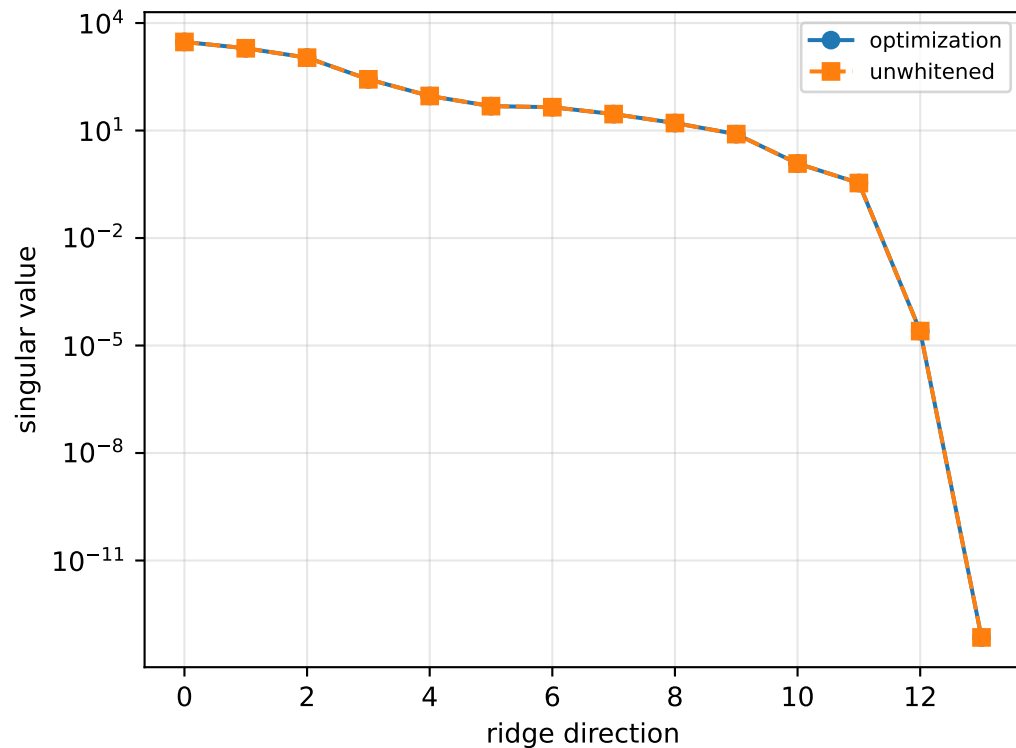
```
[[ 1.23321612e-01  1.28792241e-04 -2.47556066e-02]
 [ 1.28792241e-04  7.71409215e-02  8.41159137e-03]
 [-2.47556066e-02  8.41159137e-03  1.81055005e-01]]
```

scale-free f/m: [0.35101171 0.47075055 0.35797847 0.27489693]

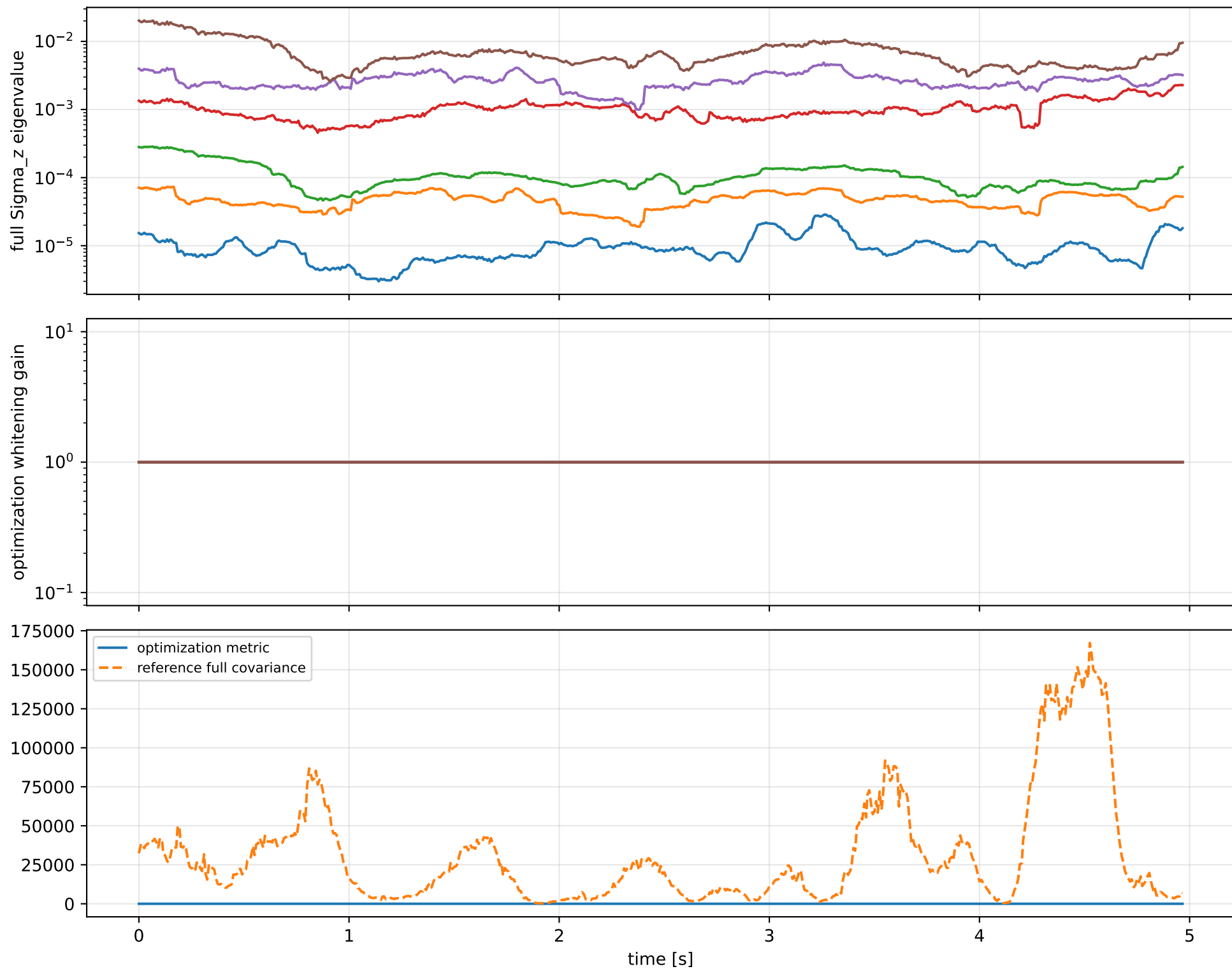
rotor lag cell: (0.218215227, 0.218225241] s

representative: 0.218220234 s

gimbal_command_replay: ridge spectrum (rank 13, nullity 1, $||\mathbf{J}\mathbf{v}||=9.13\text{e-}14$)



gimbal_command_replay: optimization vs reference covariance



gimbal_command_replay: wrench / closure (force max 1.37e-14, torque max 8.88e-16)

